A "Botany-Bot" for Digital Twin Monitoring of Occluded and Underleaf Plant Structures

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***Abstract*— Plants parts including leaves present an important example of deformable materials. We present Botany-Bot, a low-cost system for building detailed "annotated digital twins" of living plants using two stereo cameras, a digital turntable inside a lightbox, and an industrial robot arm, and 3D segmentated Gaussian Splat models. We also present robot algorithms for manipulating leaves to take high-resolution indexable images of occluded details such as stem buds and the underside/topside of leaves. Results from experiments suggest that the system can segment leaves with 90.8% accuracy, detect leaves with 86.2% accuracy, lift/push leaves with 77.9% accuracy, and take detailed overside/underside images with 77.3% accuracy.**

I. Introduction

Professional plant phenotyping platforms exist that promise a range of high-quality, high-throughput information about plants and phenotyping has the potential to improve yield and efficiency. However, these platforms are expensive [1]. Recently, emerging 3D reconstruction techniques building on Neural Radiance Fields [2] and Gaussian Splatting(GS) [3] have demonstrated impressive visual reconstruction quality [4], [5], [6] and have been used in several fields including robot grasping [7], [8], [9], [10], [11].

Plant Phenotyping Platforms. For plant phenotyping, researchers acquire height, width, mass, size, and area for various plant parts, including fruits, leaves, and stems [12]. These information allows practitioners to make informed decisions to increase plant growth, recover from pests and diseases, and maintain a healthy plant ecosystem. Traditional methods for plant phenotyping, including segmentation, used 2D information collected through sensors such as cameras and LiDAR [13]. However, 2D phenotyping often has limited performance in tracking leaf-level details due to occlusion [14]. As a result, 3D methods are increasingly favored for plant phenotyping for less occlusion [15], [16].

Robotic Plant Interaction and Motion Primitives Robot-plant interaction plays a crucial role in refining plant models and serves as a foundational step toward creating a digital twin of the plant [17]. Design of primitive robotic motions for plant interaction must account for the unique characteristics of plants, including their irregular shapes, deformable structures, and elastic properties. Existing research on plant manipulation focuses primarily on structurally destructive tasks, such as mechanical weeding [18], [19], fruit harvesting [20], and sample collection [21]. However, for

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Fig. 1: **Botany-Bot** creates detailed 3D digital twins of plants with a turntable and fixed cameras, segmenting them into individual components. Botany-Bot then augments them by lifting or pushing down individual leaves to capture high-resolution images of the occluded sides of leaves. Botany-Bot can also compute plant metrics; for example, the highlighted leaf has a leaf area of 25.6cm^2, leaf length of 7.2cm, and ground height of 27.2cm.

non-destructive interactions, such as plant inspection, gentle and precise robotic movements are essential, particularly when interacting with delicate structures like leaves. Developing such inspection primitives remains an open challenge in robotic plant interaction.

In Botany-Bot we use GS to rapidly reconstruct a model which can be viewed in High Definition(HD) at a high frame rate of 30 frame per seconds, and extend our previous work on GARField [22] to segment GS models into leaves and stems. Next, we develop an autonomous system with a 7-DOF robot arm to interact with individual leaves and take high-resolution images of their occluded detailed views (e.g. lift the individual leaf for leaf backside), capturing valuable additional detailed images for phenotyping, breeding, potential pest control, disease inspection, and growth monitoring [23], [24]. We evaluate Botany-Bot on a total of 109 leaves from 8 real plants.

II. Botany-Bot System Design

Botany-Bot introduces a plant scan system designed to capture and analyze potted plants using novel hardware and software. The system leverages a cost-effective setup featuring static cameras and a digital turntable for multi-view imaging, enabling high-quality 3D reconstructions of plants without the need for extensive camera arrays. Botany-Bot also incorporates robotic inspection that uses a detailed digital twin to autonomously examine plant features, such as leaf undersides, with precision and clarity beyond the initial reconstruction. By integrating captured images and geometric

Fig. 2: **Lightbox.** Multi-view data collection is containd within a lightbox (left) which ensures uniform directional illumination for NeRF reconstruction. (Right) a view inside the lightbox with turntable, plant, and 2 cameras.

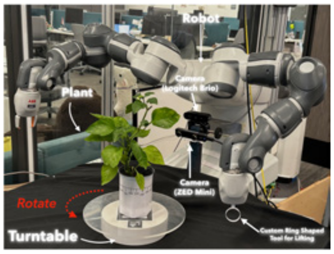

Fig. 3: **Leaf Inspection Setup.** the robot setup used for rotating the plant, lifting leaves, and imaging the undersides of plant. The robot uses a digital turntable to minimize occlusion while manipulating the plant.

data, the system offers an efficient and scalable approach to plant modeling and analysis.

A. Hardware System

1) Plant Scan System: We propose a plant scan system design using a well-lit lightbox and two fixed cameras with a digital turntable to obtain a *quasi-multiview* plant capture by rotating the potted plant around as shown in Figure 2. This design allows 3D reconstruction techniques like NeRF[2] and 3DGS [3] to obtain extensive multi-view coverage of the object being reconstructed.

2) Plant Inspection & Interaction System: The inspection system is composed of a high-resolution camera, another digital turntable to place the potted plant, and the robot arm to perform leaf manipulation equipped with a custom ring-shaped end effector to interact with the leaf while avoiding leave obstruction. Each plant is labeled with an Aruco tag, which allows registering the digital twin in the workspace of the robot. An example setup is shown in Figure 3.

B. Software System

Inside the lightbox, Botany-Bot scans a potted plant using Sec. II-A.1 to capture multi-view images. From this sequence of images, it reconstructs a detailed Gaussian Splat model and segments it into its parts including leaves, stems, pot, flowers and fruits (where available). Botany-Bot then uses this 3D model to plan leaf manipulation actions on a bimanual robot using the system in Sec. II-A.2 to closely inspect finer details that were not captured in the 3D reconstruction.

III. Methods

We assume that all plant leaves are accessible, i.e. not too densely cluttered.

A. Creating the Plant Component Data Structure

We propose the algorithm framework as follows:

1) 3D Reconstruction: We preprocess input data by automatically masking the potted plant with Segment Anything 2 (SAM 2) [25]. During radiance field construction (both NeRF and 3DGS), we do not compute standard loss functions on pixels lying outside this mask, and additionally implement a loss minimizing total rendered opacity. To alleviate lighting inconsistency due to quasi-multiview data, we surround the capture setup in a lightbox as shown in Figure 2, resulting in highly directionally uniform lighting on the rotating plant.

2) Segmentation and Leaf Detection: In addition to masking out the plant from the background to reconstruct it, we use GARField [22], to output a set of clusters corresponding to discrete groups.

Botany-Bot automatically detects leaves by analyzing the 3D gaussians within each discrete segment: we filter for the pot by rejecting the three bottommost clusters (pot, pot texture, soil), two tallest clusters (pot, stem), and clusters composed of less than 100 points (noise). The remainder of the segments are deemed a leaf.

3) Plant Modeling and Analysis: From the 3D model we automatically extract physical properties of height, number of leaves, and total leaf area.

B. Robot Primitives for Interaction and Leaf Lifting/Pushing

Although the rotating capture procedure (Sec.II-A.1) collects multi-view images of the target plant, it cannot observe key areas such as leaf undersides, leading to under-reconstruction for the digital twin. These regions are crucial for detecting issues such as pest infestations or diseases. To address this, Botany-Bot uses robot interaction with a custom end-effector to lift/push down each leaf toward a static camera, capturing high-resolution underside/overside images (Fig. 4). Guided by the plant component data structure, we solve inspection planning problem.

1) Task Primitives: The task primitive consists of three sequential steps for manipulating a target leaf using the robot arm with its inspection tool and turntable.

Rotation Alignment, Tool Positioning and Manipulation: The target leaf is rotated through the rotation of the turntable by θ such that its principal axis (i.e. the stem direction for the leaf) q_i is aligned to the high resolution camera z-axis within a small margin ϵ and the leaf center is on the camera z-axis. We denote the turntable rotation set that satisfies the alignment condition to be $\mathcal{A}_{\text{rotate}} \subset SO(2)$. The inspection tool is moved directly above or underneath the leaf center to position it for lifting/pushing. This step ensures that the tool's position p aligns with the leaf center in the horizontal plane: $\mathbf{x}_{\text{prepare}} = (x_i, y_i, z_{t_{\text{initial}}})$ where $z_{t_{\text{initial}}}$ is the initial height of the tool. The tool also needs to be parallel to the the leaf surface. We denote the inspection tool pose set that the tool positioning satisfies $\mathcal{A}_{\text{prepare}} \subset SE(3)$. The inspection tool moves upward/downward while simultaneously rotating to lift/push down the leaf. The upward motion follows: $\mathbf{x}_{\text{lift}} = (x_i, y_i, z_{t_{\text{final}}})$ where $z_{t_{\text{final}}} > z_{t_{\text{initial}}}$ while the downward motion follows: $\mathbf{x}_{\text{push}} = (x_i, y_i, z_{t_{\text{final}}})$ where $z_{t_{\text{final}}} < z_{t_{\text{initial}}}$. Simultaneously, the inspection tool rotates by an angle ϕ, applied as: $R_{\text{lift/push}} \subset SO(3)$ ensuring

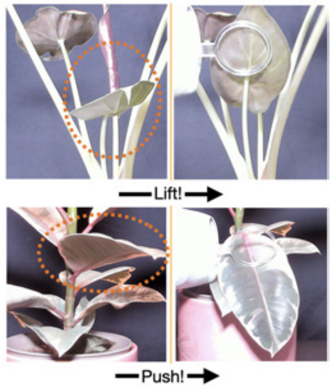

Fig. 4: **Leaf Surfaces** revealed by robot manipulation for alocasia and pp-philodendron. Each left/right pair shows a leaf before and after lifting/pushing it. Previously self-occluded regions become substantially more visible after interactive inspection.

Plant	**Leaves Segmented**	**Leaves Detected**	**MAE/MRE** (cm / %) **Length**	**Width**
Pepper 1	22/23	22/23	2.3/23.7	1.7/28.4
Pepper 2	13/17	13/17	2.4/23.5	1.2/25.6
Pepper 3	21/22	21/22	2.2/17.0	2.1/34.4
PP-Philodendron	7/7	6/7	2.4/17.5	0.8/12.4
Alocasia	6/6	5/6	3.5/24.1	1.4/17.6
Anthurium	10/12	11/12	3.0/23.9	0.9/11.3
Chroma Belize	11/13	9/13	2.2/17.5	1.0/13.0
Croton	9/9	7/9	4.0/30.3	1.6/26.9

TABLE I: **Evaluating Automatic Leaf Measurement:** For three random leaves for each plant, we estimate their length and width as described in Sec. III-A.2 (the longest and second longest lengths of the oriented bounding box extents). Note that "Leaves Segmented" only looks at the segmentation, while leaves detected looks at the true number of clusters decided to be leaves. We measure their ground-truth values using the real plants, and report the mean absolute and relative errors.

that the leaf is lifted/pushed down in a controlled manner. The inspection tool pose set $\mathcal{A}_{\text{lift/push}} \subset SE(3)$ satisfies the lifting/pushing condition. This structured sequence ensures a stable manipulation of the target leaf and we denote a feasible action set to be $\Pi = \{\pi = (a_1, a_2, a_3) : a_1 \in \mathcal{A}_{\text{rotate}}, a_2 \in \mathcal{A}_{\text{prepare}}, a_3 \in \mathcal{A}_{\text{lift}}\}$.

2) Inspection Planning: In order to obtain a high quality image observation, we propose the leaf lifting/pushing trajectory generation for inspection as an optimization problem which maximizes view coverage of the plant sub-component.

Plant	**Lifted/Pushed**	**Observed**
Pepper #1	9/15	9/9
Pepper #2	8/9	5/8
Pepper #3	8/10	2/8
PP-Philodendron	6/6	5/6
Alocasia	3/4	3/3
Anthurium	8/9	8/8
Chroma Belize*	8/9	6/8
Croton*	3/6	3/3

TABLE II: **Evaluating Botany-Bot Robot-Leaf Interactions**. We only lift/push down (*indicate push motion) leaves we find are possible by the motion planner without damaging the robot, and those that are visible in the camera already. Note that although a leaf may be lifted or moved, this does not guarantee that the leaf underside is visible to the camera.

IV. Experiments

A. Leaf Lifting/Pushing

1) Experimental Setup: We use the YuMi IRB 14000 robot arm [26] for navigating to and lifting/pushing the leaves, and a logitech Brio camera [27] for taking images of the underside/overside. Depth from a Zed mini camera[28] is used as a guide to manually align the plant pose in the world frame. For the lifting primitive, this is done once at the beginning for the sequence of manipulated leaves; while for the pushing primitive, alignment is done for every manipulated leaf. The lifting/pushing motion follows the procedure in Sec. III-B. The end effector approaches the leaf by targeting the centroid of its cluster, then lifts/pushes each leaf by 65% to 90% of the longest dimension of its oriented bounding box, proportional to its size. To evaluate Botany-Bot's interactive inspection capability, we measure the number of leaves lifted/pushed autonomously and the number of undersides/oversides fully revealed to the high-res camera. The first metric evaluates the robot's ability to interact without causing damage, while the second gauges motion accuracy.

2) Results: Table I shows the results evaluating the digital twin in segmenting, detecting and providing the length and width of leaves. Robot interaction results are reported in Table II; across 68 safely accessible leaves across 8 plants, the robot can successfully lift/push 53 leaves. Out of the 53 leaves that were pushed/lifted, their oversides/undersides are visible in 41. Example images are presented in Fig 4. The images are significantly more visually detailed, featuring the leaf veins as well as the node connecting the leaf to the rest of the stem. On the other hand, the Gaussian Splat reconstructions are under-reconstructed at the overside/underside, showing blotchy color artifacts. For observing leaf oversides/undersides, the biggest challenges are singulating the leaf and choosing the correct distance to lift/push down the leaf. Fig 1 shows an example visualization of a data structure combining Botany-Bot's physical property analyses, leaf detections, and visual observations into a single indexable model.

V. Limitations and Conclusion

The leaf lifting in Botany-Bot is currently open-loop, meaning it can fail on tightly clustered or small leaves. Incorporating more precise motion planning or closed-loop servoing for interactive leaf lifting, together with a robust physics backbone for simulating leaf motion, is a subject for future work. We present Botany-Bot, an implemented system for autonomously creating detailed 3D digital twins of plants with an inexpensive hardware setup. Botany-Bot lifts individual plant leaves with a robot to capture their undersides in high resolution, resulting in an annotated digital twin with higher visual fidelity of occluded regions. Experiments show the digital twin allows measuring of accurate physical properties, and is able to lift/push 77.9% of safely accessible and manipulatable leaves, and inspect on average 77.3% of plant leaves.

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